

Information & Entropy

- Entropy is disorder of some state or data
- Entropy is very related with thermodynamics in physics
- Information content of the image is measured in Shannon's Entropy
- In very orderly state, you do not need much space to store information about an image
- For instance, if you have an image with uniform color, all you need to store is that color and the size of the image; reducing the amount of space needed greatly
- Shannon's Entropy gives us absolute minimum amount of space that an image requires to be represented without any loss of information
- In a complete 8-bit white noise image (all values are equally likely), entropy will be 8. However, in real-life most images has much less entropy.
- When we are talking about this topic, we will not cross correlate pixels. This means, with different encoding strategies, it is possible to reduce the needed space further.

Shannon's entropy

- Calculating Shannon's entropy is easy
- First calculate PDF, then using PDF values p_i calculate the following. B is the number of bits.

$$Entropy = - \sum_{i=0}^{2^B-1} p_i \log_2(p_i)$$

- Entropy can be used to determine how much an image can be compressed. This limit is theoretical and generally cannot be matched. It is impossible to go above this limit. If you multiply this rate with 100, you will get a percentage.

$$Comp_{theoretical} = 1 - \frac{Entropy}{B}$$

Compression

- Information theory is used in image processing to compress images
- There are many methods used for compression
- Some compression methods are lossless, meaning the image is restored perfectly: PNG, bmp, tiff
- There are some methods that are lossy: JPEG
- Lossy compression methods are capable of performing compression much better, but they sacrifice image quality for it

Huffman coding

- Probably the simplest compression method is Huffman coding
- In Huffman coding each pixel value is represented with a different number of bits
- Huffman coding requires Huffman tree to work
- In compression, this tree is built
- During decompression, this tree is used
- Count of every non-zero intensity is required to build Huffman tree
- Tree is built starting from the leaves
- The lowest two non-zero entry in the histogram is joined to a node
- This node will have the pixel count of summation of left and right nodes
- Then this new node will be placed in the histogram
- This operation will continue until there is only element left in the histogram
- This last element will be the root of the tree
- Once the tree is built, starting from the root, mark left edges with 0 and right edges with 1. *Other way around works but to make my life easier follow this strategy*
- When encoding the image, replace the value of a pixel by the path indexes (0s and 1s) from the tree to get to that value
- ! An example will be provided in the class meeting
- Decompression requires bit-by-bit following the tree until a leaf is reached
- Decompression is simple and extremely fast
- This method can be used to compress any data, not just images

Run-length encoding

- In special circumstances, encoding an image in a form that first represents the intensity/color then the number of pixels that has that color can help with the compression.
- PNG algorithm has the ability to detect long chains of same color and can apply this method

Frequency domain

- Many algorithms use frequency domain to further compress images
- Both PNG and JPEG uses it
- Instead of Fourier transform, Discrete Cosine Transform (DCT) is used as it is more suitable for discrete images
- In JPEG, frequency domain transformed data is trimmed (very high frequencies are removed) and digitized
- This will cause a loss, but since it is in the high frequencies, which are generally result of noise and/or has minimal impact on the image
- However, for this purpose, JPEG compression is not suitable for computer generated images where the edges are very sharp
- Sharp edges are high frequency components and JPEG compression causes artifacts around these edges
- A very simple lossy compression algorithm can be created by applying DCT (or FFT), trimming high frequencies, and simply applying Huffman encoding to remaining values. JPEG has more rules and additional steps.